Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 1

Part 1. Give the definitions (3 points, 0.5 each)

1. Strict local minimum.
2. Positive definite and positive semidefinite matrix.
3. L-smoothness (Lipschitz gradient).
4. Lagrangian.
5. Karush–Kuhn–Tucker (KKT) conditions.
6. Dual function and dual problem.

Part 2. State and prove (3 points)

Linear convergence of GD: if f is μ-strongly convex and L-smooth, then with $\alpha = 1/L$, $f(x_k) - f^\star \le (1 - \mu/L)^k (f(x_0) - f^\star)$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 2

Part 1. Give the definitions (3 points, 0.5 each)

1. Optimization problem: decision variable, objective, feasible set.
2. Local minimum.
3. Global minimum.
4. Gradient (via the linear approximation).
5. Stationary point and saddle point.
6. Convex function.

Part 2. State and prove (3 points)

Second-order convexity criterion: $f \in C^2$ on an open convex C is convex $\iff \nabla^2 f(x) \succeq 0$ for all $x \in C$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 3

Part 1. Give the definitions (3 points, 0.5 each)

1. Directional derivative.
2. Hessian.
3. Feasible direction.
4. Convex combination and convex set.
5. Strictly convex and μ-strongly convex function.
6. Condition number.

Part 2. State and prove (3 points)

Second-order sufficient condition: if $f \in C^2$, $\nabla f(x^\star) = 0$ and $\nabla^2 f(x^\star) \succ 0$, then $x^\star$ is a strict local minimizer.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 4

Part 1. Give the definitions (3 points, 0.5 each)

1. Polyak–Lojasiewicz condition.
2. Descent direction.
3. Gradient descent with a constant step.
4. Armijo condition (backtracking).
5. Active constraint.
6. Newton’s method (Newton direction).

Part 2. State and prove (3 points)

First-order convexity criterion: a differentiable f on an open convex C is convex $\iff$ $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in C$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 5

Part 1. Give the definitions (3 points, 0.5 each)

1. Gradient (via the linear approximation).
2. Convex function.
3. Condition number.
4. Gradient descent with a constant step.
5. Armijo condition (backtracking).
6. Karush–Kuhn–Tucker (KKT) conditions.

Part 2. State and prove (3 points)

Local quadratic convergence of Newton's method: if $\nabla f(x^\star) = 0$, $\nabla^2 f(x^\star) \succ 0$ and the Hessian is M-Lipschitz, then near $x^\star$, $\|x^+ - x^\star\| \le \frac{M}{2m}\|x - x^\star\|^2$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 6

Part 1. Give the definitions (3 points, 0.5 each)

1. Strict local minimum.
2. Positive definite and positive semidefinite matrix.
3. Feasible direction.
4. Convex combination and convex set.
5. L-smoothness (Lipschitz gradient).
6. Polyak–Lojasiewicz condition.

Part 2. State and prove (3 points)

Sufficiency of the KKT conditions: if f, g_i are convex, h_j are affine and $(x^\star, \lambda^\star, \nu^\star)$ satisfies KKT, then $x^\star$ is a global minimizer.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 7

Part 1. Give the definitions (3 points, 0.5 each)

1. Directional derivative.
2. Hessian.
3. Stationary point and saddle point.
4. Strictly convex and μ-strongly convex function.
5. Active constraint.
6. Dual function and dual problem.

Part 2. State and prove (3 points)

Sufficiency of the KKT conditions: if f, g_i are convex, h_j are affine and $(x^\star, \lambda^\star, \nu^\star)$ satisfies KKT, then $x^\star$ is a global minimizer.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 8

Part 1. Give the definitions (3 points, 0.5 each)

1. Optimization problem: decision variable, objective, feasible set.
2. Local minimum.
3. Global minimum.
4. Descent direction.
5. Lagrangian.
6. Newton's method (Newton direction).

Part 2. State and prove (3 points)

Second-order sufficient condition: if $f \in C^2$, $\nabla f(x^\star) = 0$ and $\nabla^2 f(x^\star) \succ 0$, then $x^\star$ is a strict local minimizer.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 9

Part 1. Give the definitions (3 points, 0.5 each)

1. Strict local minimum.
2. Directional derivative.
3. Strictly convex and μ-strongly convex function.
4. Condition number.
5. Armijo condition (backtracking).
6. Dual function and dual problem.

Part 2. State and prove (3 points)

Sufficiency of the KKT conditions: if f, g_i are convex, h_j are affine and $(x^\star, \lambda^\star, \nu^\star)$ satisfies KKT, then $x^\star$ is a global minimizer.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 10

Part 1. Give the definitions (3 points, 0.5 each)

1. Local minimum.
2. Hessian.
3. Feasible direction.
4. Convex function.
5. Descent direction.
6. Newton’s method (Newton direction).

Part 2. State and prove (3 points)

Local quadratic convergence of Newton’s method: if $\nabla f(x^\star) = 0$, $\nabla^2 f(x^\star) \succ 0$ and the Hessian is M-Lipschitz, then near $x^\star$, $\|x^+ - x^\star\| \le \frac{M}{2m}\|x - x^\star\|^2$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 11

Part 1. Give the definitions (3 points, 0.5 each)

1. Optimization problem: decision variable, objective, feasible set.
2. Gradient (via the linear approximation).
3. Stationary point and saddle point.
4. Convex combination and convex set.
5. Lagrangian.
6. Active constraint.

Part 2. State and prove (3 points)

First-order convexity criterion: a differentiable f on an open convex C is convex $\iff$ $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in C$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 12

Part 1. Give the definitions (3 points, 0.5 each)

1. Global minimum.
2. Positive definite and positive semidefinite matrix.
3. L-smoothness (Lipschitz gradient).
4. Polyak–Lojasiewicz condition.
5. Gradient descent with a constant step.
6. Karush–Kuhn–Tucker (KKT) conditions.

Part 2. State and prove (3 points)

Linear convergence of GD: if f is μ-strongly convex and L-smooth, then with $\alpha = 1/L$, $f(x_k) - f^\star \le (1 - \mu/L)^k (f(x_0) - f^\star)$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 13

Part 1. Give the definitions (3 points, 0.5 each)

1. Stationary point and saddle point.
2. Positive definite and positive semidefinite matrix.
3. Feasible direction.
4. Descent direction.
5. Gradient descent with a constant step.
6. Karush–Kuhn–Tucker (KKT) conditions.

Part 2. State and prove (3 points)

Second-order convexity criterion: $f \in C^2$ on an open convex C is convex $\iff \nabla^2 f(x) \succeq 0$ for all $x \in C$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 14

Part 1. Give the definitions (3 points, 0.5 each)

1. Optimization problem: decision variable, objective, feasible set.
2. Strict local minimum.
3. Directional derivative.
4. Convex function.
5. Polyak–Lojasiewicz condition.
6. Active constraint.

Part 2. State and prove (3 points)

First-order convexity criterion: a differentiable f on an open convex C is convex $\iff$ $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$ for all $x, y \in C$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 15

Part 1. Give the definitions (3 points, 0.5 each)

1. Global minimum.
2. Gradient (via the linear approximation).
3. Convex combination and convex set.
4. L-smoothness (Lipschitz gradient).
5. Armijo condition (backtracking).
6. Dual function and dual problem.

Part 2. State and prove (3 points)

Second-order sufficient condition: if $f \in C^2$, $\nabla f(x^\star) = 0$ and $\nabla^2 f(x^\star) \succ 0$, then $x^\star$ is a strict local minimizer.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 16

Part 1. Give the definitions (3 points, 0.5 each)

1. Local minimum.
2. Hessian.
3. Strictly convex and μ-strongly convex function.
4. Condition number.
5. Lagrangian.
6. Newton's method (Newton direction).

Part 2. State and prove (3 points)

Linear convergence of GD: if f is μ-strongly convex and L-smooth, then with $\alpha = 1/L$, $f(x_k) - f^\star \le (1 - \mu/L)^k (f(x_0) - f^\star)$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 17

Part 1. Give the definitions (3 points, 0.5 each)

1. Gradient (via the linear approximation).
2. Stationary point and saddle point.
3. Positive definite and positive semidefinite matrix.
4. L-smoothness (Lipschitz gradient).
5. Lagrangian.
6. Newton's method (Newton direction).

Part 2. State and prove (3 points)

Descent lemma and its corollary: if ∇f is L-Lipschitz, then $f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2$; for a GD step $f(x^+) \le f(x) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f(x)\|^2$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 18

Part 1. Give the definitions (3 points, 0.5 each)

1. Strict local minimum.
2. Feasible direction.
3. Convex combination and convex set.
4. Descent direction.
5. Karush–Kuhn–Tucker (KKT) conditions.
6. Dual function and dual problem.

Part 2. State and prove (3 points)

Second-order convexity criterion: $f \in C^2$ on an open convex C is convex $\iff \nabla^2 f(x) \succeq 0$ for all $x \in C$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 19

Part 1. Give the definitions (3 points, 0.5 each)

1. Global minimum.
2. Hessian.
3. Convex function.
4. Strictly convex and μ-strongly convex function.
5. Armijo condition (backtracking).
6. Active constraint.

Part 2. State and prove (3 points)

Descent lemma and its corollary: if ∇f is L-Lipschitz, then $f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2$; for a GD step $f(x^+) \le f(x) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f(x)\|^2$.

Total: 6 points

Optimization Methods · Colloquium (Lectures 1–5)

Ticket No. 20

Part 1. Give the definitions (3 points, 0.5 each)

1. Optimization problem: decision variable, objective, feasible set.
2. Local minimum.
3. Directional derivative.
4. Polyak–Lojasiewicz condition.
5. Condition number.
6. Gradient descent with a constant step.

Part 2. State and prove (3 points)

Descent lemma and its corollary: if ∇f is L-Lipschitz, then $f(y) \le f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2$; for a GD step $f(x^+) \le f(x) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f(x)\|^2$.

Total: 6 points